

Lab 2: Web API using .NET Core with Swagger

Objective

To create an ASP.NET Core Web API with Swagger support, understand API documentation, test Web API endpoints using Swagger and Postman, and demonstrate the use of routing by customizing API endpoints through the Route attribute.

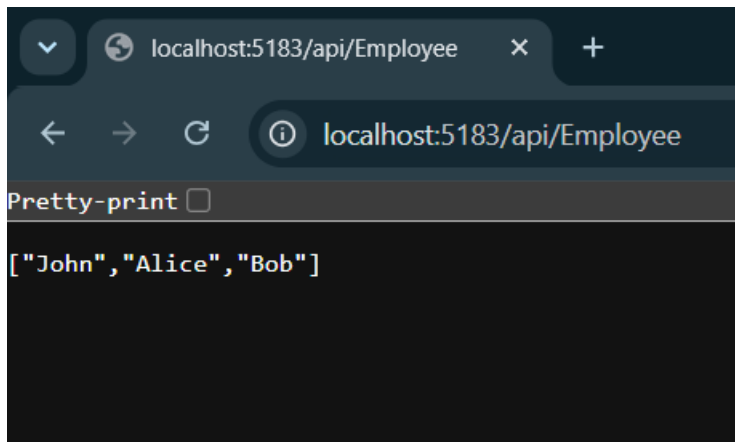
Code

Program.cs

```
Controllers > EmployeeController.cs > { } FirstWebAPI.Controllers
3  namespace FirstWebAPI.Controllers
7  public class EmployeeController : ControllerBase
9  static List<string> employees = new()
10 {
11     "John",
12     "Alice",
13     "Bob"
14 };
15
16 [HttpGet]
17 public IActionResult Get()
18 {
19     return Ok(employees);
20 }
21
22 [HttpPost]
23 public IActionResult Post(string name)
24 {
25     employees.Add(name);
26     return Ok(employees);
27 }
28
29 [HttpPut]
30 public IActionResult Put(int id, string name)
31 {
32     if (id < 0 || id >= employees.Count)
33         return BadRequest("Invalid ID");
34
35     employees[id] = name;
36     return Ok(employees);
37 }
38
39 [HttpDelete]
40 public IActionResult Delete(int id)
```

Output

```
PS C:\Users\laksh\FirstWebAPI> dotnet build
Build succeeded in 1.0s
PS C:\Users\laksh\FirstWebAPI> dotnet run
Using launch settings from C:\Users\laksh\FirstWebAPI\Properties\launchSettings.json...
Building...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5183
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\laksh\FirstWebAPI
█
```



localhost:5183/api/Employee

localhost:5183/api/Employee

Pretty-print ☐

["John","Alice","Bob"]